

Revision Notes for Class 7 Science Chapter 9 – Life Processes in Animals

All living beings need to perform certain basic activities to survive and grow. These activities, called life processes, include nutrition, respiration, excretion, circulation, and reproduction. In this chapter, we explore how animals, including humans, get their food and energy, how this food is processed, and how the body absorbs what it needs while removing wastes. A clear understanding of these life processes in animals helps us appreciate the amazing ways in which living beings maintain life.

Animals' Eating Habits

Animals exhibit a range of eating habits. For example, bees and sunbirds suck flower nectar, human infants feed on mother's milk, and snakes swallow whole prey. Aquatic animals may filter small food particles from water. All animals need food for energy, but this food is usually complex and needs to be broken down into simpler substances before usage. This process is called digestion and happens inside a special tube-like pathway called the alimentary canal.

The Human Digestive System

The alimentary canal is a long tube that starts at the mouth and ends at the anus. The journey of food starts with mechanical digestion in the mouth, where teeth break food into small pieces, and saliva softens it. Saliva also contains an enzyme that begins the process of converting starch in foods like chapati and rice into sugar, which is why prolonged chewing makes them taste sweet.

After being chewed and mixed with saliva, food is swallowed and passes down the food pipe (oesophagus) to the stomach. The walls of the oesophagus push the food downward using a wave-like movement that continues throughout the canal.

In the stomach, food mixes with digestive juices containing acid and mucus. Here, proteins begin to break down, acids kill many harmful bacteria, and mucus protects the

stomach lining. The churned food becomes semi-liquid, preparing it for further digestion.

Next, food moves to the small intestine, the longest part of the digestive system (about 6 metres). The small intestine receives bile from the liver (which breaks up fats and neutralizes acid) and pancreatic juice from the pancreas (which further digests carbohydrates, proteins, and fats). The lining of the small intestine also produces digestive juices.

Absorption and Egestion

The walls of the small intestine are lined with tiny finger-like projections called villi, which increase the surface area for absorption of digested nutrients into the blood. The blood then carries nutrients to all parts of the body for energy, growth, and repair.

Undigested food moves to the large intestine, which is wider but shorter (about 1.5 metres). Here, water and certain salts are absorbed, making the undigested material semi-solid. This waste, called stool, collects in the rectum and is expelled through the anus (egestion). Eating fibre-rich foods, like fruits and whole grains, helps the movement and removal of waste smoothly.

- Saliva starts breaking down starch to sugar in the mouth.
- Mechanical breakdown: chewing with teeth; chemical breakdown: by digestive juices.
- Stomach digests proteins and destroys bacteria with acid.
- Small intestine completes digestion and absorbs nutrients.
- Large intestine absorbs water and helps pass out waste.

Nutrition Adaptations in Animals

Not all animals digest food the same way. Grass-eating animals like cows and buffaloes are called ruminants. They chew grass briefly, swallow it, and later bring it back to the mouth to chew more thoroughly—a process called rumination. Birds do not have teeth, but instead use a muscular gizzard to grind food, often with swallowed stones. Each type of animal's digestive system has evolved to suit its preferred food.

Respiration in Humans

Animals need energy, and this is provided mainly by respiration. Respiration takes in oxygen, which helps break down glucose (sugar) in cells to produce energy, releasing carbon dioxide and water as by-products. Breathing (physical process) and respiration (chemical breakdown of food) are different but related. Humans breathe in air through the nose, and the air travels through nasal passages and the windpipe, eventually reaching tiny sacs in the lungs called alveoli.

Inside alveoli, oxygen from air passes into the blood, and carbon dioxide moves from the blood into the alveoli to be breathed out. The rib cage and diaphragm (a large muscle below the lungs) work together to expand and contract the chest, causing inhalation and exhalation. Breathing and the circulatory system (heart and blood vessels) together ensure every cell receives oxygen and nutrients, and waste is removed.

- Respiratory system parts: nostrils, nasal passages, windpipe, lungs, alveoli, rib cage, diaphragm.
- Exchange of gases occurs in alveoli: oxygen in, carbon dioxide out.
- Word equation of respiration: Glucose + Oxygen $\rightarrow$ Carbon dioxide + Water + Energy.
- Physical breathing is different from cellular respiration.

Respiration in Other Animals

Different animals have different structures for breathing. Most land animals including birds, reptiles, and mammals use lungs, though lung structure may vary. Fishes have

gills that exchange gases directly with water. Frogs use gills as tadpoles, lungs as adults, and can also breathe through moist skin. Earthworms use their skin for gas exchange. Such diversity is an adaptation to their environments.

Key Points in a Nutshell

- Life processes ensure survival and include nutrition, respiration, excretion, and more.
- Digestion occurs through coordinated action in the alimentary canal: mouth, oesophagus, stomach, small intestine, large intestine, anus, with help from liver and pancreas.
- Small intestine absorbs nutrients; large intestine absorbs water and salt.
- Ruminants chew food twice; birds have a gizzard; fish use gills; frogs and earthworms use skin or lungs depending on their stage or species.
- Breathing brings air to lungs; exchange of gases happens in alveoli.
- Respiration in cells turns nutrients and oxygen into energy, with waste products removed.
- The circulatory system delivers what is needed and takes away waste products.

Practice and Application

- To test saliva's action on starch, chewed and unchewed rice/bread can be compared with iodine solution, which turns blue-black if starch is present but does not if starch is broken into sugar.
- The diaphragm is the muscle that helps in breathing movements.
- Different structures—nostrils, windpipe, alveoli, rib cage—play key roles in the respiratory system.
- Experimenting with exhaled and inhaled air in lime water shows that exhaled air has more carbon dioxide.

Good Habits for Healthy Life Processes

- Maintain oral hygiene by regular brushing and rinsing after meals.
- Eat fibre-rich and easily digestible foods; use spices like ginger and cumin to support digestion.
- Practice mindful eating and avoid overeating.
- Avoid smoking as it harms the respiratory system.
- Breathing exercises like pranayama support lung health and well-being.
- Adapt diet and health practices if you have food-related diseases, like using millets for gluten intolerance.

Main Digestive Tract Parts and Their Roles

Digestive Part	Main Functions
Mouth	Chewing, saliva action, begins digestion of starch
Oesophagus	Moves food to stomach
Stomach	Digests proteins, churns food, uses acid and digestive juice
Small Intestine	Digests fats, carbs, proteins; absorbs nutrients
Large Intestine	Absorbs water, forms and removes stool
Liver and Pancreas	Secrete juices for further digestion

Life processes like digestion and respiration function smoothly in animals when all these body systems work in harmony and when healthy habits support them.